

RESISTANCE TO ANTIBIOTICS IN PSEUDOMONAS AND KLEBSIELLA STRAINS ISOLATED FROM POULTRY IN KARACHI

FASIHUDDIN AHMED ANSARI AND HAJRA KHATOON

Department of Microbiology, University of Karachi
Karachi 75270, Pakistan

ABSTRACT

Observations were made on the resistance of *Pseudomonas* and *Klebsiella* strains isolated from poultry in Karachi to eight commonly used antibiotics namely ampicillin (A), chloramphenicol (C), gentamycin (G), kanamycin (K), neomycin (N), polymyxin B (P), streptomycin (S) and tetracycline (T). A total of 55 strains were isolated from the liver of diseased birds after post mortem examination. The isolates were tested to determine their antibiotic resistance/sensitivity employing the above mentioned antibiotics at the levels of 50 ug/ml, 100 ug/ml & 500 ug/ml. Eighty five percent strains were found to resist one or more antibiotics in different combinations. The highest frequency of resistance was against ampicillin. One *Klebsiella* strain was found to carry two R plasmids and will be reported elsewhere.

INTRODUCTION

Poultry diseases caused by Gram negative bacteria are not uncommon (Eissa, 1981; Hofstad *et al.*, 1978). Species of *Klebsiella* and *Pseudomonas* have been reported from infected birds alongwith other pathogenic bacteria (AmirAli & Khatoon, 1976; Ansari & Khatoon 1994). Antibiotics are widely used for treating the poultry infections and as feed additives in our country as well as in other countries (Sandasovan *et al.*, 1977). Indiscriminate use of antibiotics has attributed to the widespread multiple antibiotic resistance (Goven *et al.*, 1981). The current investigation was carried out to determine the state of antibiotic resistance among *Pseudomonas* and *Klebsiella* strains isolated from poultry in Karachi using eight commonly used antibiotics. One *Klebsiella* strain was found to carry two R plasmids. Existence of R plasmid mediated antibiotic resistance in bacteria is quite significant from public health point of view as it rapidly spreads in populations (Naqvi *et al.*, 1992; Ansari & Khatoon, 1995., Khatoon *et al.*, 1998).

MATERIALS AND METHODS

Screening for antibiotic resistance

Pseudomonas and *Klebsiella* strains (total 55) screened for their antibiotic resistance were obtained from Poultry Research Institute, Korangi, Karachi. Although these strains were obtained as pure cultures, they were further purified twice on MacConkey's medium. Cultures were maintained on Tryptone agar slabs after performing various biochemical tests.

Media

The media used in the current studies included MacConkey's Agar (Difco), triple sugar iron agar (Difco), simmon citrate agar (Merck), nutrient gelatin (Merck), glucose peptone water (Merck) and urea broth (Difco).

Other media that were made in the laboratory included the following: Nutrient agar (beef extract 3gm, peptone 10 gm, NaCl 5 gm, agar agar 20 gm, distilled water 1000 ml). Luria broth (tryptone 10 gm, yeast extract 1 gm, NaCl 8 gm, distilled water 1000 ml). Tryptone agar (bacto tryptone 17 gm, agar

Table I
Antibiotic resistance patterns of *Pseudomonas* and *Klebsiella* strains
isolated from poultry in Karachi

	No. of <i>Pseudomonas</i> alongwith resistance patterns* ($\mu\text{g/ml}$) at				No. of <i>Kebsiella</i> alongwith resistance patterns* ($\mu\text{g/ml}$) at		
	50	100	500		50	100	500
ACGKNPST	7	1	-	AGKNPST	1	-	-
ACKNPST	8	7	-	AKNST	1	1	1
ACGKNPT	1	-	-	AKNT	1	1	-
ACKNPT	11	11	-	APST	1	1	-
AKNPST	1	-	-	APT	-	-	1
CKNPST	-	1	-	KNT	-	-	1
ACGPT	1	1	-	AP	1	1	-
ACKNPT	1	1	-	A	2	2	2
ACKNT	1	-	-	P	1	1	-
ACKPT	1	1	-				
ACKT	-	3	-				
ACPT	2	1	-				
ANPT	-	1	-				
ANST	1	1	-				
AKN	-	-	1				
APT	1	-	-				
AC	-	-	3				
AK	-	-	3				
AN	1	1	-				
AS	-	-	1				
NS	-	-	1				
PS	-	-	1				
PT	-	1	-				
ST	1	1	-				
A	-	-	23				
P	-	-	2				
T	1	1	-				

A = ampicillin
N = neomycin

C = chloramphenicol
P = polymyxin B

G = gentamycin
S = streptomycin

K = kanamycin
T = tetracycline

agar 6 gm, distilled water 1000 ml pH 7). MacConkey's agar, Nutrient agar and Luria broth were used for the growth of bacteria, whereas, other media were used for characterization.

Antibiotics

The antibiotics used were: ampicillin trihydrate, chloramphenicol (both from Opal laboratories Ltd., Karachi), gentamycin (from Aspro Nicholas Ltd., Karachi), kanamycin sulphate (from Continental Pharma, Belgium), neomycin sulphate (from Glaxo Laboratories, Karachi), polymyxin B sulphate, streptomycin sulphate (both from Pfizer Laboratories,

Karachi) and tetracycline hydrochloride (from Lederle Laboratories, Karachi). All the antibiotics were obtained in purified powder form and stock solutions (10 mg/ml) were made in distilled water. Chloramphenicol was dissolved in absolute alcohol. The solutions were sterilized by millipore filtration and kept frozen if not in use. The solutions of daily use were kept refrigerated at 4°C for not more than a week.

Antibiotic resistance/sensitivity of the stains

Screening for antibiotic resistance/sensitivity was performed by the methods described by Khursheed and Khatoun (1984).

RESULTS AND DISCUSSION

Pseudomonas and *Klebsiella* strains isolated from poultry in Karachi were screened for their resistance to the commonly used antibiotics: ampicillin, chloramphenicol, gentamycin, kanamycin, neomycin, polymyxin B, streptomycin and tetracycline at the levels of 50 µg/ml, 100 µg/ml and 500 µg/ml. Of the 55 strains screened, the number of resistant bacteria at the three increasing levels were 48, 41 and 40 respectively. Antibiotic resistance patterns of these strains at the three levels can be examined in Table I.

From the results presented here, it appears that multiple antibiotic resistance among *Pseudomonas* and *Klebsiella* isolated from poultry in Karachi is not uncommon as out of 55 strains screened for resistance, 85% were resistant to one or more antibiotics at 50 µg/ml, 74% were resistant to one or more antibiotics at 100 µg/ml and 72 % were resistant to one or more antibiotics at 500 µg/ml (Table I). The resistance at as high as 500 µg/ml is alarming because if bacteria become resistant to such a high level of antibiotics, disease treatment with antibiotics would not be possible.

Several combination of antibiotic resistance (resistance patterns) form *Pseudomonas* and *Klebsiella* strains isolated from poultry in Karachi were observed at the three levels of antibiotics (Table I). The number of resistance patterns at the levels of 50 µg/ml, 100 µg/ml and 500 µg/ml were 22, 21 and 12 respectively (Table I). The most common pattern for *Pseudomonas* at 50 µg/ml and 100 µg/ml was ACKNPT, whereas at 500 µg/ml, the most common pattern was A. The most common pattern for *Klebsiella* was A at all the levels of antibiotics used for screening.

In case of *Pseudomonas* the highest frequency of resistance was against ampicillin at all the levels of antibiotics used for screening, whereas, the lowest frequency of

resistance was against gentamycin. In case of *Klebsiella*, all the strains were sensitive to chloramphenicol and it seems to be the drug of choice for treating infections caused by this organism.

ACKNOWLEDGEMENTS

The authors are thankful to the Director and staff of the Poultry Research Institute, Korangi, Karachi for providing bacterial isolates and some literature. The donation of antibiotics in purified powder from various pharmaceutical laboratories is also appreciated. The work described here is partly supported by research grant from Dean, Faculty of Science, University of Karachi.

REFERENCES

- AmirAlis S. and Khatoon H. (1976). *J. Sc. Kar. Univ.* 3:84.
- Ansari F. A. and Khatoon H. (1994). *Indian J. Exp. Biol.* 32: 211.
- Ansari F.A. and Khatoon H. (1995). *Pakistan J. Zool.* 27(4): 375-377.
- Eissa Y. M. (1981). Prevalent bacterial diseases of poultry. In Handbook on poultry diseases prevalent in Pakistan. Poultry Research Institute, Karachi., pp 26-35.
- Goren E., de Jong W. A. and van Solkema, A (1981). *Avian Pathol.* 10: 43-55.
- Hofstad M.S., Calnek, B.W., Helmboldt, C.F., Reid W. M. and Yoder Jr, H.W. (1978). Diseases of poultry. Iowa State University Press., U.S.A.
- Khatoon H., Saeed, A. and Naseem S. (1998). *J. Sc. Kar. Univ.* 26(2): 37-45.
- Khursheed B. and Khatoon H. (1984). *Kar. Univ. J. Sc.* 12: 223-230.
- Naqvi S.Z.H., Ansari F.A. and Khatoon H. (1992). *Pakistan J. Zool.* 22(2): 141-147.
- Sandasovan P.R., Srinivasan V.A., Venugopalan, A.T. and Balaprakasam R.A. (1977). *Avian Diseases.* 21: 136-138.